Consider you are provided with a database with 3 tables, that holds data from an Industrial manufacturing plant. You need to perform data preprocessing on the provided data. The tables are:

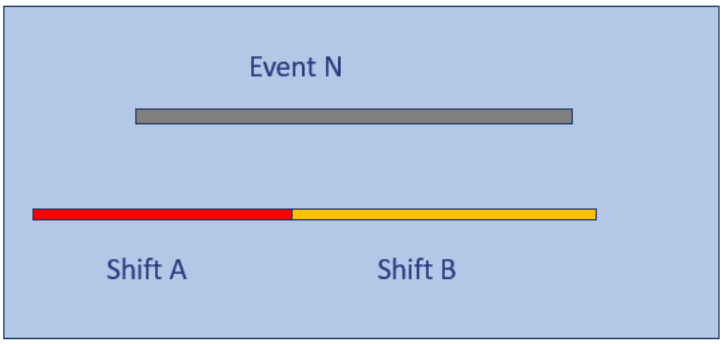
1. Event Table: Holds the events generated by different IOT devices functioning in the manufacturing plant.
2. Shift Table: Holds the shift details of the IOT devices in the plant. The devices used in the plant function in 3 shifts (A, B & C) per day. The A shift begins at 6 AM and ends at 2:30 PM, B shift begins at 2:30 PM and ends at 11 PM and C shift begins at 11 PM and ends at next day 6 AM.
3. Break Table: Holds details of the breaks provided with in the plant and breaks happened to the IOT devices.

You task is to perform the below mentioned data cleaning.

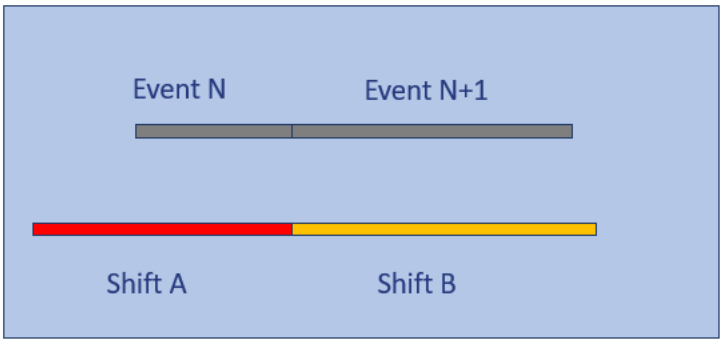
1. **Cut of events with change of shift**

If an event is spread across one or more shift, than that event should be split.

Before



After



In this case event N is spread across SHIFT A and SHFT B, so this event should be split into two. An example is given below.

Before:

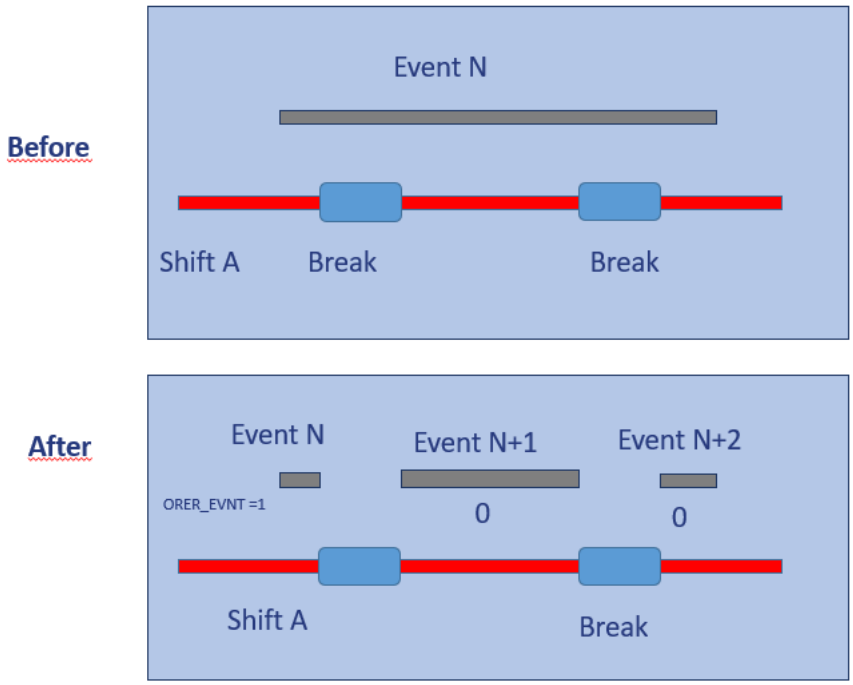
device_id	shift_id	shift_code	event_id	event_start_time	event_end_time
18717	18717#A#2023-10-17 06:00	A	989	2023-10-17T14:28:23.343Z	2023-10-17T14:35:04.79Z

After:

device_id	shift_id	shift_code	event_id	event_start_time	event_end_time
18717	18717#A#2023-10-17 06:00	A	989	2023-10-17T14:28:23	2023-10-17T14:30:00
18717	18717#B#2023-10-17 06:00	B	989	2023-10-17T14:30:00	2023-10-17T14:35:04

2. **Cut of events for breaks inside shifts**

   If an event spread across a break inside a shift that should be cut during the break period.



In this case, event N is spread across 2 breaks in the shift A, so that event is cut off. An example is given below.

Before:

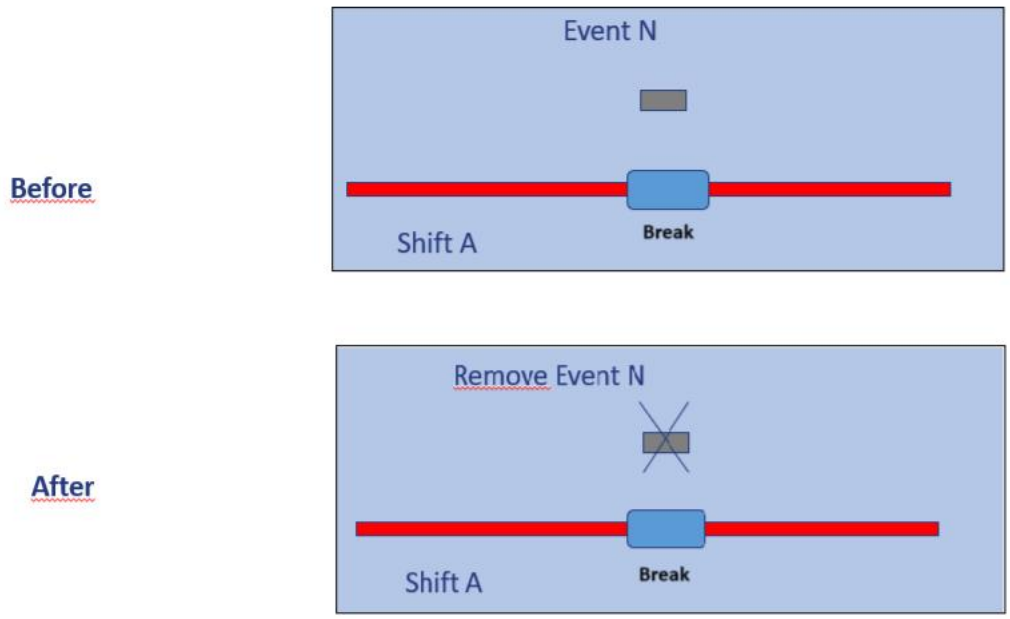
device_id	shift_id	event_id	event_start_time	event_end_time	break_start_time	break_end_time
18717	18717#A#2023-10-17 06:00	989	2023-10-17T10:28:23	2023-10-17T10:50:00	2023-10-17T10:30:00	2023-10-17T10:35:00

After:

device_id	shift_id	event_id	event_start_time	event_end_time	break_start_time	break_end_time
18717	18717#A#2023-10-17 06:00	989	2023-10-17T10:28:23	2023-10-17T10:30:00	2023-10-17T10:30:00	2023-10-17T10:35:00
18717	18717#A#2023-10-17 06:00	989	2023-10-17T10:35:00	2023-10-17T10:50:00	2023-10-17T10:30:00	2023-10-17T10:35:00

3. **Remove of events during break**

Remove of events occurred during a break period.



In this case the event N occurred during a break period, so removed that event.

Note:

- Use MYSQL database for uploading the data provided along with this and perform the data preprocessing. MYSQL can be downloaded from this URL https://www.mysql.com/downloads/ .
- Once done, share the scripts for reviewing.

Once you submit your response on the deadline, we will review your content and get on call with you.